COMPUTER NETWORKS AND CLOUD COMPUTING

Comprehensive Study Notes

Topics 1–10 · Weightage: 10%

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Data Communication

1.1 What is Data Communication?

Data communication refers to the exchange of data between two devices via a transmission medium. It involves the transfer of data (text, numbers, images, audio, video) from one point to another in a disciplined and orderly way.

Five Components of Data Communication

- **Message** – The information (data) to be communicated.
- **Sender** – The device that sends the data message (computer, phone, camera).
- **Receiver** – The device that receives the message.
- **Transmission Medium** – The physical path by which a message travels (cable, air).
- **Protocol** – A set of rules that govern data communication (TCP/IP, HTTP).

1.2 Data Flow Directions

Mode	Description
Simplex	Data flows in ONE direction only. Example: keyboard to computer, TV broadcast.
Half-Duplex	Data flows in BOTH directions but only ONE at a time. Example: walkie-talkie.
Full-Duplex	Data flows in BOTH directions SIMULTANEOUSLY. Example: telephone call.

1.3 Transmission Modes

Serial Transmission

Bits are sent one after another over a single channel. Slower but cheap and simple. Used in long-distance communication.

Parallel Transmission

Multiple bits are sent simultaneously over multiple channels. Faster but expensive. Used in short-distance (e.g., printer cables).

1.4 Analog vs Digital Data

Property	Analog	Digital
Nature	Continuous signal	Discrete (0s and 1s)
Bandwidth	Lower	Higher

Noise Immunity	Low	High
Example	AM/FM radio, vinyl records	Computer data, CDs
Transmission	Susceptible to noise	Easily regenerated/amplified

1.5 Bandwidth and Data Rate

Bandwidth: The range of frequencies a medium can carry. Measured in Hertz (Hz). Higher bandwidth = more data capacity.

Data Rate (Bit Rate): Number of bits transmitted per second (bps). Common units: Kbps, Mbps, Gbps.

Nyquist Theorem (Noiseless Channel): Maximum data rate = $2 \times B \times \log_2(V)$, where B = bandwidth, V = signal levels.

Shannon's Theorem (Noisy Channel): Maximum data rate = $B \times \log_2(1 + SNR)$, where SNR = Signal-to-Noise Ratio.

■ *Shannon's Theorem gives the theoretical maximum; real-world rates are always lower.*

1.6 Transmission Impairments

- **Attenuation:** Loss of energy as signal travels through a medium. Solved using amplifiers/repeaters.
- **Distortion:** Change in the shape of the signal. Occurs when different frequencies travel at different speeds.
- **Noise:** Unwanted signals added to the data. Types: Thermal, Induced, Crosstalk, Impulse noise.

Computer Networks

2.1 What is a Computer Network?

A computer network is a collection of interconnected devices (computers, printers, servers) that share resources and communicate with each other. Networks enable resource sharing, communication, and centralized data management.

Advantages of Computer Networks

- Resource sharing (printers, files, internet)
- Communication (email, chat, VoIP)
- Centralized data storage and backup
- Cost reduction by sharing hardware
- Remote access and collaboration

2.2 Network Classifications by Size

Type	Full Form	Range	Example
PAN	Personal Area Network	~10 m	Bluetooth, USB connections between personal devices
LAN	Local Area Network	~1 km	Office or building network, Ethernet, Wi-Fi
MAN	Metropolitan Area Network	~100 km	City-wide network, cable TV networks
WAN	Wide Area Network	Global	Internet, satellite links, MPLS

2.3 Network Topologies

Bus Topology

All devices share a single communication line. Simple and cheap but a single break disables the whole network.

Star Topology

All devices connect to a central hub/switch. Easy to manage; hub failure disables all. Most common in LANs.

Ring Topology

Devices form a closed loop. Data travels in one direction. A single failure can disrupt the network.

Mesh Topology

Every device connects to every other device. Highly reliable but expensive and complex.

Hybrid Topology

Combination of two or more topologies. Most real-world networks use this.

Tree Topology

Hierarchical structure combining bus and star. Used in wide-area networks.

2.4 Network Models

OSI Model (7 Layers)

Layer	Function & Protocols
7 – Application	User interface, HTTP, FTP, SMTP, DNS
6 – Presentation	Encryption, compression, data translation (SSL/TLS)
5 – Session	Session management, synchronization, checkpoints
4 – Transport	End-to-end delivery, TCP/UDP, flow control, error recovery
3 – Network	Routing, IP addressing, packet forwarding
2 – Data Link	Framing, MAC addressing, error detection (CRC)
1 – Physical	Bit transmission, cables, connectors, voltages

2.5 Network Devices

Device	Description
Hub	Layer 1 device. Broadcasts data to all ports. No intelligence.
Switch	Layer 2 device. Forwards data only to the destination port using MAC table.
Router	Layer 3 device. Routes packets between different networks using IP addresses.
Bridge	Connects two LAN segments. Filters traffic based on MAC address.
Gateway	Connects networks with different protocols. Often includes routing and NAT.
Modem	Modulates/demodulates signals for transmission over telephone lines.
Repeater	Amplifies/regenerates signals to extend network distance.
Access Point	Provides wireless connectivity (Wi-Fi) to wired network.

Data Link Layer

3.1 Overview & Responsibilities

The Data Link Layer (Layer 2 of OSI) is responsible for node-to-node delivery of data frames. It takes raw bits from the Physical Layer and organizes them into frames, ensuring reliable transmission over a single link.

Key Functions of Data Link Layer

- **Framing:** Encapsulates packets into frames with header/trailer.
- **Physical Addressing:** Uses MAC addresses to identify sender and receiver.
- **Flow Control:** Prevents sender from overwhelming receiver.
- **Error Control:** Detects and retransmits lost or corrupted frames.
- **Access Control:** Determines which device has control of the channel.

3.2 Framing

Framing divides the stream of bits into manageable data units called frames. Methods include:

- **Fixed-size framing:** All frames are the same size (ATM cells = 53 bytes).
- **Variable-size framing:** Frames can vary in size. Needs delimiter flags.
- **Character stuffing:** Escape characters added before delimiter characters in data.
- **Bit stuffing:** A 0-bit is inserted after five consecutive 1-bits in the data.

3.3 Error Detection & Correction

Parity Check

A parity bit is added to make total number of 1s even (even parity) or odd (odd parity). Can detect single-bit errors only.

Cyclic Redundancy Check (CRC)

More powerful error detection using polynomial division. A CRC value is appended to the frame. Receiver recalculates CRC; if it doesn't match, an error occurred. Used in Ethernet.

Hamming Code

Error-correcting code that can detect and correct single-bit errors. Uses multiple parity bits placed at power-of-2 positions.

3.4 Flow Control Protocols

Stop-and-Wait

Sender sends one frame and waits for ACK before sending the next. Simple but inefficient (poor utilization).

Sliding Window

Multiple frames can be sent before requiring ACK. Window size determines how many unacknowledged frames can be outstanding.

- **Go-Back-N ARQ:** Receiver only accepts in-order frames. On error, sender retransmits from the errored frame onwards.
- **Selective Repeat ARQ:** Only the erroneous frame is retransmitted. More efficient but needs larger buffers.

3.5 Multiple Access Protocols

When multiple devices share the same channel, a protocol is needed to manage access.

Protocol	Description
TDMA	Time Division Multiple Access – each station gets a time slot.
FDMA	Frequency Division – each station gets a frequency band.
CSMA/CD	Carrier Sense Multiple Access / Collision Detection – used in Ethernet. Stations listen before transmitting.
CSMA/CA	Collision Avoidance – used in Wi-Fi (802.11). Avoids collisions using RTS/CTS mechanism.
ALOHA	Random access: stations transmit whenever ready. Pure ALOHA and Slotted ALOHA variants.

3.6 MAC Address & Ethernet

A **MAC (Media Access Control) address** is a 48-bit hardware address (e.g., 00:1A:2B:3C:4D:5E) burned into network interface cards. The first 24 bits identify the manufacturer (OUI) and the last 24 bits are device-specific.

Ethernet uses CSMA/CD for access control. IEEE 802.3 standard. Frame includes preamble, destination MAC, source MAC, type/length, data, and CRC.

Network Layer

4.1 Role of the Network Layer

The Network Layer (Layer 3) is responsible for logical addressing and routing of packets from source to destination across multiple networks. Unlike the Data Link Layer which handles local delivery, the Network Layer provides end-to-end delivery.

Core Responsibilities

- **Logical Addressing:** Uses IP addresses to identify hosts across networks.
- **Routing:** Determines the best path for packets to travel.
- **Packet Forwarding:** Moves packets from input to output port in a router.
- **Fragmentation & Reassembly:** Splits large packets for transmission, reassembles at destination.
- **Internetworking:** Connects heterogeneous networks using routers.

4.2 IP Addressing (IPv4)

An IPv4 address is a 32-bit number written as four octets in dotted decimal notation (e.g., 192.168.1.10). It has two parts: **Network ID** and **Host ID**.

IP Address Classes

Class	Range	Default Mask	Usage
A	0.0.0.0 – 127.255.255.255	/8	Large organizations, ISPs
B	128.0.0.0 – 191.255.255.255	/16	Medium organizations
C	192.0.0.0 – 223.255.255.255	/24	Small networks (most common)
D	224.0.0.0 – 239.255.255.255	N/A	Multicasting
E	240.0.0.0 – 255.255.255.255	N/A	Research/reserved

4.3 Subnetting & CIDR

Subnetting divides a large network into smaller sub-networks to improve performance and security. A subnet mask (e.g., 255.255.255.0) defines which part of the IP address is the network portion.

CIDR (Classless Inter-Domain Routing): Uses slash notation (e.g., 192.168.1.0/24). More flexible than classful addressing. Eliminates class boundaries.

■ *Subnet formula: Number of subnets = 2^n (n = borrowed bits); Hosts per subnet = 2^h – 2 (h = remaining host bits)*

4.4 IPv6

IPv6 is the next-generation IP protocol with a 128-bit address space (e.g., 2001:0db8:85a3::8a2e:0370:7334). Provides ~3.4×10^{38} unique addresses.

IPv6 Features

- 128-bit address space — solves IPv4 exhaustion
- Auto-configuration (SLAAC) — no DHCP needed
- Built-in IPSec for security
- Simplified header (40 bytes fixed)
- No broadcast — uses multicast and anycast instead
- Better QoS with flow label field

4.5 Routing Algorithms

Distance Vector Routing (RIP)

Each router maintains a table of distances to all destinations. Updates are shared with neighbors periodically. Simple but slow to converge (count-to-infinity problem).

Link State Routing (OSPF)

Each router has complete knowledge of the network topology. Uses Dijkstra's shortest path algorithm. Fast convergence, scalable.

Path Vector Routing (BGP)

Used in the Internet between autonomous systems. Each router stores the complete path to each destination. Policy-based routing decisions.

Transport Layer

5.1 Role of the Transport Layer

The Transport Layer (Layer 4) provides end-to-end communication services for applications. It acts as the intermediary between the upper-layer applications and the lower-layer network protocols.

Key Services

- **Process-to-Process Communication:** Uses port numbers to identify applications.
- **Segmentation & Reassembly:** Divides application data into segments.
- **Connection Control:** Can be connection-oriented (TCP) or connectionless (UDP).
- **Flow Control:** Prevents sender from overwhelming receiver.
- **Error Control:** Ensures segments arrive correctly and in order.
- **Multiplexing/Demultiplexing:** Multiple applications share the same network.

5.2 TCP – Transmission Control Protocol

TCP is a **connection-oriented, reliable** transport protocol. It guarantees delivery, ordering, and error-checking.

TCP Three-Way Handshake (Connection Establishment)

- Step 1 – **SYN:** Client sends SYN (synchronize) packet with initial sequence number.
- Step 2 – **SYN-ACK:** Server replies with SYN-ACK, acknowledging and sending its sequence number.
- Step 3 – **ACK:** Client sends ACK. Connection is established.

TCP Connection Termination (Four-Way)

- FIN → ACK → FIN → ACK sequence used to gracefully close connection.

TCP Header Fields (Key)

Field	Purpose
Source/Dest Port	16-bit port numbers identifying applications (e.g., HTTP=80, HTTPS=443)
Sequence Number	32-bit number identifying position of data in byte stream
Acknowledgment No.	Next byte expected from sender
Window Size	Number of bytes receiver is willing to accept (flow control)
Flags (SYN/ACK/FIN)	Control bits for connection management
Checksum	Error detection for header and data

5.3 UDP – User Datagram Protocol

UDP is a **connectionless, unreliable** protocol. No connection setup, no guarantee of delivery or ordering. Minimal overhead makes it fast.

UDP Use Cases

- DNS – Domain Name System lookups
- DHCP – IP address assignment
- VoIP and video streaming (speed > reliability)
- Online gaming (low latency critical)
- TFTP – Trivial File Transfer Protocol
- SNMP – Network management

5.4 TCP vs UDP Comparison

Feature	TCP	UDP
Connection	Connection-oriented (handshake)	Connectionless
Reliability	Guaranteed delivery	No guarantee
Ordering	In-order delivery	No ordering
Speed	Slower (overhead)	Faster (minimal overhead)
Flow Control	Yes (windowing)	No
Error Control	Yes (retransmission)	Checksum only
Header Size	20–60 bytes	8 bytes
Examples	HTTP, FTP, SMTP, SSH	DNS, DHCP, VoIP, Gaming

5.5 Port Numbers & Sockets

A **port number** (0–65535) identifies a specific application/process. A **socket** = IP address + Port number (e.g., 192.168.1.10:80).

- **Well-known ports:** 0–1023 (HTTP=80, HTTPS=443, FTP=21, SSH=22, SMTP=25, DNS=53)
- **Registered ports:** 1024–49151 (user/server applications)
- **Dynamic/Ephemeral ports:** 49152–65535 (temporary client ports)

Application Layer

6.1 Overview

The Application Layer (Layer 7) is the topmost layer, providing network services directly to user applications. It defines the protocols and interfaces that applications use to communicate over a network.

6.2 DNS – Domain Name System

DNS translates human-readable domain names (e.g., www.google.com) into IP addresses. It uses a hierarchical distributed database.

DNS Hierarchy

- **Root Level (.):** Top of the hierarchy, 13 root server clusters globally.
- **Top-Level Domain (TLD):** .com, .org, .net, .edu, .pk, .uk, etc.
- **Second-Level Domain:** google.com, microsoft.com, etc.
- **Subdomain:** mail.google.com, www.google.com

DNS Resolution Process (Recursive)

Client → Local DNS → Root DNS → TLD DNS → Authoritative DNS → IP returned to client. Results are cached with TTL values.

6.3 HTTP & HTTPS

HTTP (HyperText Transfer Protocol) is the foundation of web communication. Works on request-response model. Uses TCP port 80. HTTPS adds TLS/SSL encryption, uses port 443.

HTTP Method	Purpose
GET	Retrieve data from server (fetching a webpage)
POST	Send data to server (submitting a form)
PUT	Update existing resource on server
DELETE	Remove a resource from server
HEAD	Retrieve headers only (no body)

6.4 Email Protocols

SMTP (Port 25/587)

Simple Mail Transfer Protocol – sends email from client to server and between servers.

POP3 (Port 110)

Post Office Protocol v3 – downloads mail to local device and deletes from server.

IMAP (Port 143)

Internet Message Access Protocol – reads email on server; stays synchronized across devices.

6.5 FTP – File Transfer Protocol

FTP (Port 21) transfers files between client and server. Uses two connections: control (port 21) and data (port 20). SFTP (Secure FTP) uses SSH for encryption. FTPS uses SSL/TLS.

6.6 Other Key Application Protocols

Protocol	Port	Function
SSH	22	Secure remote login and command execution
Telnet	23	Unsecured remote login (deprecated for SSH)
DHCP	67/68	Automatically assigns IP addresses to hosts
SNMP	161	Network monitoring and management
NTP	123	Synchronizes clocks across network devices
RDP	3389	Remote Desktop Protocol for Windows
LDAP	389	Directory services (Active Directory)

Wireless Networks

7.1 Introduction to Wireless Networks

Wireless networks transmit data over radio waves, infrared, or microwave signals without physical cables. They offer mobility and flexibility but face challenges in security, interference, and signal degradation.

7.2 Wi-Fi (IEEE 802.11 Standards)

Standard	Year	Freq	Max Speed	Key Feature
802.11a	1999	5 GHz	54 Mbps	Short range
802.11b	1999	2.4 GHz	11 Mbps	Wider range, interference
802.11g	2003	2.4 GHz	54 Mbps	Backward compatible
802.11n	2009	2.4/5 GHz	600 Mbps	MIMO technology
802.11ac	2013	5 GHz	~3.5 Gbps	MU-MIMO, beamforming
802.11ax (Wi-Fi 6)	2019	2.4/5/6 GHz	~9.6 Gbps	OFDMA, dense environments

7.3 Bluetooth & Other Short-Range Technologies

Bluetooth (IEEE 802.15.1)

Range ~10-100m, 2.4 GHz. Used for personal devices. Versions: BT 4.0 (BLE), BT 5.0 (2x faster, 4x range).

Zigbee (IEEE 802.15.4)

Low power, low data rate. Used in IoT, smart home devices. Range ~10-100m.

NFC

Near Field Communication. Range <10cm. Used in contactless payments, access cards.

RFID

Radio Frequency Identification. Used in inventory tracking, passports, access control.

IrDA

Infrared Data Association. Line-of-sight communication. Used in TV remotes.

7.4 Cellular Networks

Era & Capabilities	
	Analog voice only (AMPS)

	Digital voice, SMS (GSM, CDMA)
	Mobile internet, video calls (~2 Mbps)
	High-speed internet (~100 Mbps), VoIP
	Ultra-low latency, ~1-10 Gbps, IoT, autonomous vehicles

7.5 Wireless Network Architecture

Infrastructure Mode

Devices connect through an Access Point (AP). AP connects to wired network. Most common setup in homes and businesses.

Ad-Hoc Mode (IBSS)

Devices connect directly to each other without an AP. Peer-to-peer. Used for temporary networks.

Mesh Network

Multiple APs create overlapping coverage. Packets can route through multiple APs. Used for large coverage areas.

7.6 Wireless Security

- **WEP:** Wired Equivalent Privacy – deprecated, easily cracked.
- **WPA:** Wi-Fi Protected Access – uses TKIP, improved over WEP.
- **WPA2:** Uses AES-CCMP encryption. Strong security, widely used.
- **WPA3:** Latest standard. SAE (Simultaneous Authentication of Equals), protection against offline attacks.
- **802.1X:** Enterprise authentication using RADIUS server.

Cloud Computing

8.1 What is Cloud Computing?

Cloud computing is the delivery of computing services (servers, storage, databases, networking, software, analytics) over the internet ('the cloud') to offer faster innovation, flexible resources, and economies of scale. Users pay only for what they use.

NIST Definition – Five Essential Characteristics

- **On-demand self-service:** Provision resources without human interaction.
- **Broad network access:** Services available over standard network interfaces.
- **Resource pooling:** Multi-tenant model with dynamic resource assignment.
- **Rapid elasticity:** Capabilities scale up/down quickly and automatically.
- **Measured service:** Usage is monitored, controlled, and reported.

8.2 Cloud Service Models

IaaS – Infrastructure as a Service

Provides virtualized computing infrastructure over the internet. User manages OS, middleware, applications. Provider manages physical infrastructure.

- **Examples:** AWS EC2, Microsoft Azure VMs, Google Compute Engine

PaaS – Platform as a Service

Provides a platform for developers to build, deploy, and manage applications without managing underlying infrastructure.

- **Examples:** Google App Engine, AWS Elastic Beanstalk, Heroku

SaaS – Software as a Service

Fully functional software delivered over the internet. No installation or maintenance required.

- **Examples:** Gmail, Microsoft 365, Salesforce, Dropbox, Zoom

8.3 Cloud Deployment Models

Deployment Model	Description
Public Cloud	Owned by third-party provider. Resources shared among customers. Cost-effective, scalable. (AWS, Azure, GCP)
Private Cloud	Dedicated to a single organization. Higher security and control. More expensive. (OpenStack, VMware)

Hybrid Cloud	Combines public and private clouds. Sensitive data in private, scalable workloads in public.
Community Cloud	Shared among organizations with common concerns (e.g., government agencies, healthcare).
Multi-Cloud	Uses services from multiple cloud providers to avoid vendor lock-in.

8.4 Cloud Providers & Services

Amazon Web Services (AWS)

Market leader. Services: EC2 (compute), S3 (storage), RDS (database), Lambda (serverless), VPC (networking), Route 53 (DNS), CloudFront (CDN).

Microsoft Azure

Strong enterprise integration. Services: Azure VMs, Azure Blob Storage, Azure SQL, Azure Functions, Active Directory, Azure DevOps.

Google Cloud Platform (GCP)

Strong in AI/ML and data analytics. Services: GCE, GCS, BigQuery, Cloud Spanner, Kubernetes Engine (GKE).

8.5 Cloud Concepts

Virtualization

Creating virtual versions of servers, storage, and networks using hypervisors (VMware, Hyper-V, KVM).

Containerization

Lightweight virtualization using containers (Docker, Kubernetes). Share OS kernel.

Serverless Computing

Function as a Service (FaaS) – run code without managing servers. (AWS Lambda, Azure Functions)

Auto-Scaling

Automatically adjust resources based on demand. Cost-efficient and high availability.

CDN

Content Delivery Network – caches content at edge locations globally to reduce latency.

Load Balancing

Distributes incoming traffic across multiple servers for high availability.

Cloud Challenges

- **Security & Privacy:** Data breaches, unauthorized access, compliance.
- **Downtime:** Dependence on provider's uptime (SLA guarantees).
- **Vendor Lock-in:** Difficulty migrating away from one provider.
- **Limited Control:** No control over underlying infrastructure.
- **Cost Management:** Unexpected costs if resources not monitored.

Network Security (Networks Perspective)

9.1 Introduction to Network Security

Network security involves policies, practices, and tools to prevent, detect, and respond to unauthorized access, misuse, modification, or denial of computer networks. The CIA Triad forms the foundation:

CIA Triad

- **Confidentiality:** Ensuring information is only accessible to authorized parties (encryption).
- **Integrity:** Ensuring data is accurate and hasn't been tampered with (hashing, digital signatures).
- **Availability:** Ensuring systems are operational and accessible when needed (redundancy, DDoS protection).

9.2 Common Network Attacks

Attack Type	Description & Mitigation
DoS / DDoS	Denial of Service – overwhelms a server/network with traffic. DDoS uses multiple sources (botnet). Mitigation: rate limiting, CDN, firewalls.
Man-in-the-Middle (MitM)	Attacker intercepts communication between two parties. Can eavesdrop or alter data. Mitigation: TLS/SSL, certificate pinning.
SQL Injection	Malicious SQL code inserted into input fields to manipulate database. Mitigation: parameterized queries, input validation.
Phishing	Fraudulent emails/websites trick users into revealing credentials. Mitigation: user education, MFA, email filters.
ARP Spoofing	Attacker sends fake ARP messages to link their MAC with a legitimate IP. Enables MitM on local network.
DNS Spoofing	Corrupts DNS cache to redirect users to malicious sites. Mitigation: DNSSEC.
Packet Sniffing	Capturing network packets to steal sensitive data. Mitigation: encryption, VPN, HTTPS.
Port Scanning	Discovering open ports on a host. Mitigation: firewall rules, intrusion detection.
Ransomware	Malware that encrypts files and demands ransom. Mitigation: backups, security patches, antivirus.
Zero-Day Attack	Exploits unknown vulnerability before patch is available. Mitigation: behavior-based detection, network segmentation.

9.3 Firewalls

A firewall monitors and controls incoming/outgoing network traffic based on predetermined security rules. Acts as a barrier between trusted and untrusted networks.

- **Packet Filter Firewall:** Inspects packets at network layer; filters by IP, port, protocol.
- **Stateful Firewall:** Tracks connection state; allows only established connections.
- **Application/Proxy Firewall:** Inspects application layer; understands HTTP, FTP traffic.
- **NGFW (Next-Gen Firewall):** Deep packet inspection, IDS/IPS, SSL inspection, application awareness.

9.4 Cryptography & Encryption

Symmetric Encryption

Same key used for encryption and decryption. Fast. Key distribution is a challenge. Examples: AES (128/256-bit), DES, 3DES, RC4.

Asymmetric Encryption (Public Key)

Uses a key pair: public key (encrypt) and private key (decrypt). Slow but solves key distribution. Examples: RSA, ECC, Diffie-Hellman.

Hashing

One-way function; converts data to fixed-size hash. Used for password storage, integrity verification. Examples: MD5 (deprecated), SHA-1 (deprecated), SHA-256, SHA-3.

9.5 Security Protocols

Protocol	Description
SSL/TLS	Encrypts data in transit (HTTPS). TLS 1.3 is current standard.
IPSec	Secures IP packets. Used in VPNs. Two modes: Transport and Tunnel.
SSH	Secure remote access and file transfer. Replaces Telnet and FTP.
HTTPS	HTTP over TLS/SSL. Port 443. Secure web browsing.
WPA3	Latest Wi-Fi security standard using SAE authentication.
DNSSEC	Adds cryptographic signatures to DNS responses to prevent spoofing.
802.1X	Port-based network access control with RADIUS authentication.
VPN	Creates encrypted tunnel over public network. Types: IPSec, SSL, PPTP.

9.6 IDS & IPS

- **IDS (Intrusion Detection System):** Monitors network traffic and alerts on suspicious activity. Passive – does not block.
- **IPS (Intrusion Prevention System):** Actively blocks detected threats in real-time. Inline with traffic.

- **SIEM:** Security Information and Event Management – centralizes logging and analysis from multiple sources.

Next Generation Networks

10.1 What are Next Generation Networks?

Next Generation Networks (NGN) refer to a broad class of evolving network technologies that replace traditional circuit-switched telephony with packet-switched networks. NGN integrates voice, data, and multimedia over a unified IP-based infrastructure.

NGN Key Characteristics

- Packet-based transport (IP/MPLS)
- Separation of transport and service layers
- Broadband capabilities with QoS support
- Support for multiple broadband technologies
- Unrestricted access to different service providers
- Support for generalized mobility (seamless handoff)

10.2 Software Defined Networking (SDN)

SDN decouples the network's control plane (decision-making) from the data plane (packet forwarding). A centralized controller manages network behavior through software.

SDN Architecture

- **Application Layer:** Network applications (load balancer, firewall, QoS).
- **Control Layer:** SDN Controller (OpenFlow, ONOS, OpenDaylight). The 'brain'.
- **Infrastructure Layer:** Network devices (switches, routers) – only forward packets as instructed.

SDN Benefits

- Centralized management and visibility
- Programmable networks – rapid deployment
- Vendor-agnostic hardware
- Dynamic traffic engineering
- Reduced operational costs

10.3 Network Function Virtualization (NFV)

NFV virtualizes network functions (firewalls, load balancers, routers, IDS/IPS) that traditionally run on dedicated hardware, deploying them as software on commodity servers.

- **VNF (Virtual Network Function):** Virtualized version of a network function.
- **MANO (Management & Orchestration):** Manages NFV infrastructure and VNF lifecycle.
- **Benefits:** Reduced CAPEX/OPEX, flexibility, faster service deployment.

10.4 5G Networks

5G is the fifth generation mobile network standard, designed to provide ultra-fast speeds, ultra-low latency, and massive connectivity for IoT.

5G Key Features

- **eMBB:** Enhanced Mobile Broadband – peak speeds up to 20 Gbps.
- **URLLC:** Ultra-Reliable Low Latency Communications – latency as low as 1ms.
- **mMTC:** Massive Machine Type Communications – supports millions of IoT devices/km^2.
- **Network Slicing:** Multiple virtual networks on a single physical 5G infrastructure.
- **Beamforming:** Directional signal transmission for better coverage and speed.
- **Massive MIMO:** Hundreds of antennas for improved spectral efficiency.

10.5 Internet of Things (IoT)

IoT refers to the network of physical devices embedded with sensors, software, and connectivity, enabling them to collect and exchange data.

- **IoT Protocols:** MQTT, CoAP, AMQP, Zigbee, Z-Wave, LoRaWAN.
- **Edge Computing:** Processing data near the source rather than in the cloud. Reduces latency.
- **Fog Computing:** Extends cloud computing to the edge. Intermediate layer between IoT devices and cloud.

10.6 Emerging Technologies

Technology	Description
Intent-Based Networking (IBN)	Network automatically configures itself based on high-level business intent. Uses AI/ML.
Quantum Networking	Uses quantum entanglement and superposition for theoretically unbreakable encryption (QKD).
Li-Fi	Light Fidelity – uses visible light for wireless communication. Up to 100x faster than Wi-Fi.
Named Data Networking (NDN)	Content-centric networking – routes based on content name rather than IP address.
White Box Networking	Using open-source software on commodity hardware instead of proprietary network devices.
Network Automation & AIOps	Using AI/ML to automate network management, anomaly detection, and optimization.